

BATCH RECORD for *E. coli* GROWTH in BioFlo 120 BIOREACTOR

BioFlo 120 BIOREACTOR CONTROL STATION SERIAL # _____

GROUP ID: _____ Group A GROUP MEMBERS (INITIALS): PA PB PC PD PE PF PG PH PI PJ

	Date (DD/MM/YY)	Time Point when sample is taken (e.g. t = 0 min)	Clock Time when sample is taken (Use Military Time)	Agitation (RPM) (bioreactor's screen)	Temperature (°C) (bioreactor's screen)	pH Reading (bioreactor's screen)	pH Measured (pH meter)	Gas Flow (SLPM) (bioreactor's screen)	Notes
Sample 1	02/10/26	t = 0 min	0945	250	37.2	6.79	6.65	1.0	N/A
	Operator/Verifier PE /PI	Operator/Verifier PE /PI	Operator/Verifier PE /PI	Operator/Verifier PE /PI	Operator/Verifier PE /PI	Operator/Verifier PE /PI	Operator/Verifier PE /PI	Operator/Verifier PE /PI	Operator/Verifier PE /PI
Sample 2	02/10/26	T=46 min	1032	250	37.0	6.66	6.53	1.0	N/A
	Operator/Verifier PI /PE	Operator/Verifier PE /PI	Operator/Verifier PI /PE	Operator/Verifier PI /PE	Operator/Verifier PI /PE	Operator/Verifier PI /PE	Operator/Verifier PI /PE	Operator/Verifier PI /PE	Operator/Verifier PI /PE
Sample 3	02/10/26	T=97 min	1123	250	37.0	6.40	6.19	1.0	N/A
	Operator/Verifier PJ /PH	Operator/Verifier PJ PK /PH	Operator/Verifier PJ PK /PH	Operator/Verifier PJ PL /PH	Operator/Verifier PJ PK /PH	Operator/Verifier PJ PK /PH	Operator/Verifier PJ PK /PH	Operator/Verifier PJ PK /PH	Operator/Verifier PJ /PH
Sample 4	02/10/26	T=143 min	1209	250	37.0	5.91	5.63	1.0	N/A
	Operator/Verifier PH /PJ PK	Operator/Verifier PH /PJ PK	Operator/Verifier PH /PJ PK	Operator/Verifier PH /PJ PK	Operator/Verifier PH /PJ PK	Operator/Verifier PH /PJ PK	Operator/Verifier PH /PJ PK	Operator/Verifier /PJ PK	Operator/Verifier PH /PJ PK
Sample 5	02/10/26	T=190 min	1256	250	37.0	5.57	5.29	1.0	N/A
	Operator/Verifier PC /PF	Operator/Verifier PC /PF	Operator/Verifier PC /PF	Operator/Verifier PC /PF	Operator/Verifier PC /PF	Operator/Verifier PC /PF	Operator/Verifier PC /PF	Operator/Verifier PC /PF	Operator/Verifier PC /PF
Sample 6	02/10/26	T=240 min	1346	250	37.0	5.43	5.21	1.0	N/A
	Operator/Verifier PF /PC	Operator/Verifier PF /PC	Operator/Verifier PF /PC	Operator/Verifier PF /PC	Operator/Verifier PF /PC	Operator/Verifier PF /PC	Operator/Verifier PF /PC	Operator/Verifier PF /PC	Operator/Verifier PF /PC

*Operator and Verifier must put initials

BATCH RECORD for *E. coli* GROWTH in BioFlo 120 BIOREACTOR

NOTES:

BATCH RECORD for *E. coli* GROWTH in BioFlo 120 BIOREACTOR

	Date (DD/MM/YY)	Time Point when sample is taken (e.g. t = 0 min)	Clock Time when sample is taken (Use Military Time)	Agitation (RPM) (bioreactor's screen)	Temperature (°C) (bioreactor's screen)	pH Reading (bioreactor's screen)	pH Measured (pH meter)	Gas Flow (SLPM) (bioreactor's screen)	Notes
Sample 7	02/10/26	T=287 min	1433	250	37.0	5.38	5.22	1.0	N/A
	Operator/Verifier PB / PA	Operator/Verifier PB / PA	Operator/Verifier PB / PA	Operator/Verifier PB / PA	Operator/Verifier PB / PA	Operator/Verifier PB / PA	Operator/Verifier PB / PA	Operator/Verifier PB / PA	Operator/Verifier PB / PA
Sample 8	02/10/26	T=332 min	1518	250	37.0	5.38	5.23	1.0	N/A
	Operator/Verifier PA / PB	Operator/Verifier PA / PB	Operator/Verifier PA / PB	Operator/Verifier PA / PB	Operator/Verifier PA / PB	Operator/Verifier PA / PB	Operator/Verifier PA / PB	Operator/Verifier PA / PB	Operator/Verifier PA / PB
Sample 9	2/10/26	T= 361 min	1547	250	37.0	5.4	5.21	1.0	N/A
	Operator/Verifier PD / PA	Operator/Verifier PD / PA	Operator/Verifier PD / PA	Operator/Verifier PD / PA	Operator/Verifier PD / PA	Operator/Verifier PD / PA	Operator/Verifier PD / PA	Operator/Verifier PD / PA	Operator/Verifier PD / PA
Sample 10									
	Operator/Verifier /	Operator/Verifier /	Operator/Verifier /	Operator/Verifier /	Operator/Verifier /	Operator/Verifier /	Operator/Verifier /	Operator/Verifier /	Operator/Verifier /
Sample 11									
	Operator/Verifier /	Operator/Verifier /	Operator/Verifier /	Operator/Verifier /	Operator/Verifier /	Operator/Verifier /	Operator/Verifier /	Operator/Verifier /	Operator/Verifier /
Sample 12									
	Operator/Verifier /	Operator/Verifier /	Operator/Verifier /	Operator/Verifier /	Operator/Verifier /	Operator/Verifier /	Operator/Verifier /	Operator/Verifier /	Operator/Verifier /

*Operator and Verifier must put initials

BATCH RECORD for *E. coli* GROWTH in BioFlo 120 BIOREACTOR

Sampling #	Microcentrifuge Tube Weight (g)	Weight of Microcentrifuge Tube with Pellet (g)	Calculated Pellet Weight (g)	Quadrant Streak	Gram Stain Performed (Yes / No)	Gram Staining Observations	Comments
1	1.5267g	1.5375	0.0108	yeah	Y / N	Very few cells seen; difficult to determine	N/A
	Operator/Verifier PA / PB	Operator/Verifier PA / PB	Operator/Verifier PA / PB	Operator/Verifier PC / PF	Operator/Verifier PH / PJ , PK	Operator/Verifier PH / PJ , PK	Operator/Verifier PH / PJ , PK
2	1.5375g	1.407	0.0032	Performed	Y / N	More cells visible clear rod-shaped	N/A
	Operator/Verifier PB / PA	Operator/Verifier PB / PA	Operator/Verifier PB / PA	Operator/Verifier PF / PC	Operator/Verifier PJ / PH , PK	Operator/Verifier PJ / PH , PK	Operator/Verifier PJ / PH , PK
3	1.5521	1.5645	0.0174	Completed	Y / N	Multiple, purple rod-shaped bacteria visualized	N/A
	Operator/Verifier PD / PG	Operator/Verifier PD / PG	Operator/Verifier PD / PG	Operator/Verifier PB / PA	Operator/Verifier PF / PC	Operator/Verifier PF / PC	Operator/Verifier PF / PC

*Operator and Verifier must put initials

BATCH RECORD for *E. coli* GROWTH in BioFlo 120 BIOREACTOR

Sampling #	Microcentrifuge Tube Weight (g)	Weight of Microcentrifuge Tube with Pellet (g)	Calculated Pellet Weight (g)	Quadrant Streak	Gram Stain Performed (Yes / No)	Gram Staining Observations	Comments
4	1.5410	1.5509	0.0099	Executed	Y / N	Bacteria were there	N/A
	Operator/Verifier PI /PG	Operator/Verifier PI /PG	Operator/Verifier PI /PG	Operator/Verifier PA /PB	Operator/Verifier PC /PF	Operator/Verifier PC /PF	Operator/Verifier PC /PF
5	1.5284	1.5327	0.0043	Done & Dusted	Y / N	Rod-shaped bacteria seen spread out, or in small clumps	N/A
	Operator/Verifier PE /PI	Operator/Verifier PE /PI	Operator/Verifier PE /PI	Operator/Verifier PG /PD	Operator/Verifier PA /PB	Operator/Verifier PA /PB	Operator/Verifier PA /PB
6	1.5395	1.5447	0.0052	Over	Y / N	Small, pink rod- shaped bacteria spread about	N/A
	Operator/Verifier PI /PE	Operator/Verifier PI /PE	Operator/Verifier PI /PE	Operator/Verifier PD /PG	Operator/Verifier PB /PA	Operator/Verifier PB /PA	Operator/Verifier PB /PA

*Operator and Verifier must put initials

BATCH RECORD for E. coli GROWTH in BioFlo 120 BIOREACTOR

Sampling #	Microcentrifuge Tube Weight (g)	Weight of Microcentrifuge Tube with Pellet (g)	Calculated Pellet Weight (g)	Quadrant Streak	Gram Stain Performed (Yes / No)	Gram Staining Observations	Comments
7	1.5414g	1.5488	0.0074	Accomplished	Y / N	Too much cells, Clearly visible rod shaped	N/A
	Operator/Verifier PJ PK /PH	Operator/Verifier PJ PK /PH	Operator/Verifier PJ PK /PH	Operator/Verifier PI /PE	Operator/Verifier PG /PD	Operator/Verifier PG /PD	Operator/Verifier PG /PD
8	1.5493	1.5768	0.0275	Adequately Executed	Y / N	Cells clearly visible, but may have washed some away	N/A
	Operator/Verifier PH /PK PJ	Operator/Verifier PH /PJ	Operator/Verifier PH /PJ	Operator/Verifier PE /PI	Operator/Verifier PD /PG	Operator/Verifier PD /PG	Operator/Verifier PD /PG
9	1.5484	1.5574	0.015	Successful	Y / N	Rod shaped pink bacterium in clusters, obvious growth	Had to do twice, ran out of saffron
	Operator/Verifier	Operator/Verifier	Operator/Verifier	Operator/Verifier	Operator/Verifier	Operator/Verifier	Operator/Verifier

BATCH RECORD for E. coli GROWTH in BioFlo 120 BIOREACTOR

Sampling #	Microcentrifuge Tube Weight (g)	Weight of Microcentrifuge Tube with Pellet (g)	Calculated Pellet Weight (g)	Quadrant Streak	Gram Stain Performed (Yes / No)	Gram Staining Observations	Comments
	PF /PC	PF /PC	PF /PC	PH /PJ PK	PE /PI	PE /PI	PE /PI
10	*Operator and Verifier must put initials				Y / N		
	Operator/Verifier /	Operator/Verifier /	Operator/Verifier /	Operator/Verifier /	Operator/Verifier /	Operator/Verifier /	Operator/Verifier /
11					Y / N		
	Operator/Verifier /	Operator/Verifier /	Operator/Verifier /	Operator/Verifier /	Operator/Verifier /	Operator/Verifier /	Operator/Verifier /
12					Y / N		

BATCH RECORD for E. coli GROWTH in BioFlo 120 BIOREACTOR

Sampling #	Microcentrifuge Tube Weight (g)	Weight of Microcentrifuge Tube with Pellet (g)	Calculated Pellet Weight (g)	Quadrant Streak	Gram Stain Performed (Yes / No)	Gram Staining Observations	Comments
	Operator/Verifier /	Operator/Verifier /	Operator/Verifier /	Operator/Verifier /	Operator/Verifier /	Operator/Verifier /	Operator/Verifier /

*Operator and Verifier must put initials

NOTES:

BATCH RECORD for *E. coli* GROWTH in BioFlo 120 BIOREACTOR

	Date (DD/MM/YY)	Time Point when sample is taken. (e.g. t = 0 min)	Absorbance (A) of Undiluted Sample		Dilution Factor (DF) if absorbance ≥ 1.0 au (DF= 1 if no dilution performed)		Corrected Absorbance (A) @600nm $A = DF * A_{600}$ (If dilution is performed, A_{600} is the absorbance of the diluted sample.)		Notes
Sample 1	02/10/26	t = 0 min	Sample #1 0.172	Sample #2 N/A	Sample #1 1	Sample #2 N/A	Sample #1 N/A	Sample #2	
	Operator/Verifier PD / PG	Operator/Verifier PD / PG	Operator/Verifier PD / PG		Operator/Verifier PD / PG		Operator/Verifier PD / PG		
Sample 2	02/10/26	T=46 min	Sample #1 0.359	Sample #2 N/A	Sample #1 1	Sample #2 N/A	Sample #1 N/A	Sample #2 N/A	
	Operator/Verifier PD / PG	Operator/Verifier PI / PE	Operator/Verifier PG / PD		Operator/Verifier PI / PE		Operator/Verifier PI / PE		
Sample 3	02/10/26	T=91 min	Sample #1 0.640	Sample #2 N/A	Sample #1 1	Sample #2 N/A	Sample #1 N/A	Sample #2 N/A	
	Operator/Verifier PI / PE	Operator/Verifier PI / PE	Operator/Verifier PI / PE		Operator/Verifier PI / PE		Operator/Verifier PI / PE		
Sample 4	02/10/26	T=143 min	Sample #1 0.863	Sample #2 N/A	Sample #1 1	Sample #2 N/A	Sample #1 N/A	Sample #2 N/A	
	Operator/Verifier PG / PI	Operator/Verifier PG / PI	Operator/Verifier PG / PI		Operator/Verifier PG / PI		Operator/Verifier PG / PI		
Sample 5	02/10/26	T=190 min	Sample #1 0.972	Sample #2 N/A	Sample #1 1	Sample #2 N/A	Sample #1 N/A	Sample #2 N/A	
	Operator/Verifier PH / PJ PK	Operator/Verifier PH / PJ PK	Operator/Verifier PH / PJ PK		Operator/Verifier PH / PJ PK		Operator/Verifier PH / PJ PK		
Sample 6	02/10/26	T=240 min	Sample #1 0.996	Sample #2 N/A	Sample #1 1	Sample #2 N/A	Sample #1 N/A	Sample #2 N/A	
	Operator/Verifier PJ PK / PH	Operator/Verifier PJ PK / PH	Operator/Verifier PJ PK / PH		Operator/Verifier PJ PK / PH		Operator/Verifier PJ PK / PH		

*Operator and Verifier must put initials

** Two separate samples will be measured for absorbance.

BATCH RECORD for *E. coli* GROWTH in BioFlo 120 BIOREACTOR

	Date (DD/MM/YY)	Time Point when sample is taken. (e.g. t = 0 min)	Absorbance (A) @600nm of Undiluted Sample		Dilution Factor (DF) if absorbance ≥ 1.0 au (DF= 1 if no dilution performed)		Corrected Absorbance (A) A= DF * A ₆₀₀ (If dilution is performed, A ₆₀₀ is the absorbance of the diluted sample.)		Notes
Sample 7	02/10/26	T= 287 min	Sample #1 1.057	Sample #2 N/A	Sample #1 2	Sample #2 N/A	Sample #1 1.320	Sample #2 N/A	Diluted with 0.5mL of LB media 0.660 x 2 = 1.320
	Operator/Verifier PF /PC	Operator/Verifier PF /PC	Operator/Verifier PF /PC		Operator/Verifier PF /PC		Operator/Verifier PF /PC		
Sample 8	02/10/26	T= 332	Sample #1 1.054	Sample #2 N/A	Sample #1 2	Sample #2 N/A	Sample #1 1.404	Sample #2 N/A	Diluted with 0.5mL of LB media 0.702 x 2 = 1.404
	Operator/Verifier PC /PF	Operator/Verifier PC /PF	Operator/Verifier PC /PF		Operator/Verifier PC /PF		Operator/Verifier PC /PF		
Sample 9	02/10/26	T=361	Sample #1 1.078	Sample #2 N/A	Sample #1 2	Sample #2 N/A	Sample #1 1.374	Sample #2 N/A	Diluted with 0.5mL of LB media 0.687 x 2 = 1.374
	Operator/Verifier PA /PB	Operator/Verifier PA /PB	Operator/Verifier PA /PB		Operator/Verifier PA /PB		Operator/Verifier PA /PB		
Sample 10			Sample #1	Sample #2 N/A	Sample #1	Sample #2	Sample #1	Sample #2	
	Operator/Verifier /	Operator/Verifier /	Operator/Verifier /		Operator/Verifier /		Operator/Verifier /		
Sample 11			Sample #1	Sample #2 N/A	Sample #1	Sample #2	Sample #1	Sample #2	
	Operator/Verifier /	Operator/Verifier /	Operator/Verifier /		Operator/Verifier /		Operator/Verifier /		
Sample 12			Sample #1	Sample #2 N/A	Sample #1	Sample #2	Sample #1	Sample #2	
	Operator/Verifier /	Operator/Verifier /	Operator/Verifier /		Operator/Verifier /		Operator/Verifier /		

*Operator and Verifier must put initials

** Two separate samples will be measured for absorbance.